

2D MATRIX PHASED ARRAY SPECIFICATIONS

Piezocomposite Technology



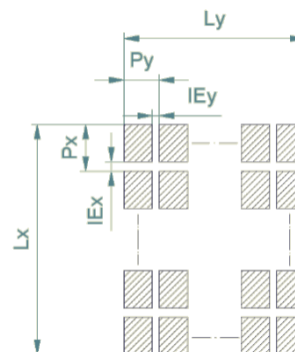
CUSTOMER reference: 5MHz 8X8 elts
IMASONIC reference: CDC16278-1

CONTEXT

Objective is to manufacture identical to Imasonic probe produce under referent T108P , but with acoustical matching to rexolite and flanges on lateral sides of the casing

ACOUSTICAL CHARACTERISTICS AND ACCEPTANCE LEVEL

Array type: 2-D Matrix array
Number of channels ($N_x * N_y$): 8X8 elements
Mechanical focusing: none, flat active area
Pitch in primary axis (P_x): 0.6 mm
Inter element spacing (IE_x): 0.1 mm
Pitch in secondary axis (P_y): 0.8 mm
Inter element spacing (IE_y): 0.1 mm
Total active area ($L_x * L_y$): 4.7 x 6.3 mm²



Acoustic design ☒ Versatile ☐ Accuracy ☐ Hard face ☐ Custom
Centre frequency (-6dB): 5 MHz $\pm$ 10% ⁽¹⁾ ⁽²⁾
Acoustical impedance matching: Rexolite (2.45 MRayl)

Pulse Duration (-20dB): $\leq$ 700 ns ⁽¹⁾ ⁽²⁾
Bandwidth (-6dB): $\geq$ 50 % ⁽¹⁾ ⁽²⁾

Homogeneity in sensitivity $\pm$ 3 dB ⁽¹⁾
Number of non-working elements: 0 maximum (adjacent or not)

These elements will not be taken into account in the determination of the mean values of the parameters.

⁽¹⁾ Measurement conditions described in performances measurement and test paragraph

⁽²⁾ Average value calculated on all the elements

MECHANICAL CHARACTERISTICS

Material: Stainless steel
Geometry: }
Dimensions: } According to general drawing T108P 03 001 00
Particular tolerances: }
Cable output: }
Probe holder interface: } Flanges on small sides for wedge fixation, dimensions to be discussed at the order

An overall drawing will be sent for approval before manufacturing

INTERCONNECTIONS

Cable: 50 Ω Multi-coaxial with overall shielding
Protective jacket: none
Length: 3.0 m $\pm$ 0.1 m
Connectors: Ipex
Pinout diagram: standard

Other specification according to ***“Linear phased array probes standard specifications-revision 7”*** attached